

A Network Model of HIV Transmission

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Abstract

We developed a graph-based simulation model of HIV transmission dynamics using individual viral load tracking and geographic placement of populations and clinics. The model incorporates close-contact and proximity-based transmission, ART initiation, and viral load progression over weekly time steps. Preliminary results demonstrate expected epidemiological patterns, including density-dependent spread and nonlinear relationships between clinic access and mortality. The simulation provides a flexible framework for exploring public health interventions.

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1 Introduction

HIV remains a large, global problem. According to United Nations statistics, there have been a total of more than 80 million HIV cases since the beginning of the AIDS epidemic, resulting in more than 40 million deaths. Over 40 million people were still living with HIV in 2025, with over 1 million new cases and 570,000 estimated deaths from HIV-related causes in 2025. AIDS deaths remained high in 2025, indicating that the epidemic is still ongoing. We sought to build a data-based model of AIDS to help guide public health decisions.

The datasets used in this study were derived from real clinical studies and population surveys. Our model works by modeling individuals as nodes in a graph, using the NetworkX package as a base for graph operations. This approach allows us to track connections between people as well as how those connections chain through each other; for example, A is connected to B, who is connected to C, etc. HIV infection is more complex than a binary state. The viral load of an individual refers to the copies of the virus that can be expected to be found in a milliliter of blood; a higher viral load (VL) means the infection is more liable to spread and poses a greater danger to the individual. By individually tracking each simulated person's viral load, we can control more parameters and bring the simulation closer to a useful reflection of reality.

This paper will specifically focus on HIV spread in sub-Saharan Africa, due to the availability and focus of treatment in this region of the world.

2 Methods

2.1 Data

For a project that aims to model real world-behavior on a large scale, first-party data collection is not feasible, as this would require a large-scale study or survey on HIV. Instead, we cite several clinic studies, surveys, and meta-analysis papers concerned with HIV in sub-Saharan Africa. Specifically, citations of works such as The epidemiology of HIV population viral load in twelve sub-Saharan African countries [1] and Prevalence and determinants of HIV among reproductive-age women (15–49-years) in Africa from 2010 to 2019: a multilevel analysis of demographic and health survey data [2] provide demographic and transmission statistics. One-year mortality among adults with advanced HIV in sub-Saharan Africa: a systematic review and meta-analysis [4] and Baseline characteristics associated with mortality among children living with HIV initiating ART at a rural district HIV clinic of Mozambique [5] provide information on mortality statistics. Modelling Disease Spread through Random and Regular Contacts in Clustered Populations [7], Modelling HIV/AIDS Epidemiological Complexity: A Scoping Review of Agent-Based Models and Their Application [8], and Bayesian inference of set-point viral load transmission models [9] provide additional reference for HIV progression patterns.

2.2 Simulation Setup

We set up our population according to specific parameters and then ran a series of "ticks" in the simulation, each representing a week in real time. Each tick of the simulation calculated the infections that occur from each node to each of their neighbors, calculated the mortality rate of any nodes that have severe untreated viral levels, and then calculated the progression of medical efforts to place some individuals on medical treatment for HIV called Anti-Retroviral Therapy (ART).

Setting up the simulation requires several parameters. We considered two primary factors: geographical location and "close contacts." Geographical location was used to represent a small likelihood of proximal spread, while close contacts were used to represent connections that represent a likely vector of spread for HIV. For geographical location, we generated a number of "cities" around which the population on the graph would center. These were generated uniformly at random on a coordinate plane of fixed height and width. These "city center locations" were placed randomly around the map, and then each individual node in the graph was placed into one of these random cities, distributed some distance from the center according to a user-specified distribution (for which this study defaults to a standard t-distribution, although a normal

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distribution could also be considered). Then, each node was either selected to begin the simulation infected or non-infected, according to a user-specified proportion which is set to 0.01 for this project.

2.3 Mortality

To tick the graph, we followed several steps. First, we determined any deaths due to unsuppressed viral load by plugging the viral load for an infected individual into a chained exponential logistic function. This was done to mirror individual death rates: the year-to-year death rate for a completely unsuppressed viral load meant that the death rate at a completely unchecked viral load should only be 5% for a weekly tick. The function used is

$$f(v) = \frac{0.05}{1 + e^{-2(e^{0.0002v} - 3.5)}} \quad (1)$$

2.4 Infection

Next, each infected node checked its infection probability for each other node in the graph. Each node checked all nodes that it had a close connection edge with (which had no connection with geographic distance) and generated a probability based on the viral load of the infected individual. This viral load infection function was a logistic curve, with low infection probability for low viral load, nearing 0 at what is considered "undetectable" viral load levels of 50 or lower. The probability of infection was generated with the following equation:

$$f(v) = \frac{1}{1 + e^{-0.1(v-250)}} \quad (2)$$

Another check was then made for proximal, distance-based infection, with approximately a 20% lower rate, varying depending on the distance between the two nodes. This utilizes a smooth bump function shown below as an additional factor of simulated infection probability:

$$f(d) = \begin{cases} e^{\frac{1}{(\frac{d}{r})^2 - 1}}, & d < 1 \\ 0, & d \geq 1 \end{cases} \quad (3)$$

where r is the maximum proximal infection radius. Our simulation defaults this to 1.

Older versions of the simulation had performed this check between every two nodes of the graph, meaning the runtime for each simulation would scale quadratically with the number of nodes in the graph. We implemented a caching system whereby the simulation remembers the nodes that have a nonzero chance of proximal spread between them, allowing the simulation to scale linearly with the number of nodes in the simulation at the cost of some overhead time to compute relevant proximal neighbors and additional space for storing proximal neighbors. The condition for being considered a proximal neighbor is having a proximal infection factor of greater than 0.01, meaning that the chance of proximal infection is at least 1% as likely as the standard infection probability.

2.5 Viral Load Progression

The simulation then checked to change viral load based on whether each individual was on ART or not, a characteristic stored for each node. For nodes on ART, this was done using an exponential decay function, where the viral load would drop very quickly if viral load

was high and would slowly settle at an undetectable level below 50 copies per mL. For this, we use the function

$$\frac{dV}{dt} = -0.6(V - 45) \quad (4)$$

by which each node changes its existing viral load by the given amount each tick if it is on ART.

For unsuppressed individuals without ART, this was done by using the derivative of a logistic function that modeled untreated viral load progression and modifying viral load by that amount. Viral load increases quickly and then reaches a maximum, so we modeled it using a logistic function. The function computes its current position on the curve

$$f(x) = \frac{1000000}{1 + e^{-3(week-3)}} \quad (5)$$

using an inverse function and then sets the viral load to the value

$$f(x) = \frac{1000000}{1 + e^{-3((week+1)-3)}} \quad (6)$$

2.6 ART Initiation Clinics

Finally, the simulation checked to initiate some nodes for ART. During simulation setup, a specified number of clinics were placed in geographic locations following similar rules to individuals. Each tick, an infected node had a certain chance to initiate ART based on its distance to a clinic. This used the same bump function as proximal infection chances and could be tweaked to specify maximum chance of initiation as well as effective radius of these clinics.

The initiation chance is given by

$$a(d) = \begin{cases} e^{\frac{m}{(d/r)^2 - 1}}, & d < r \\ 0, & d \geq r \end{cases} \quad (7)$$

where r is the effective clinic radius, m is the maximum possible chance for a person to initiate ART if they have 0 distance from a clinic, and both are modifiable by the user. We default to $r = 1$, $m = 0.5$.

3 Results

For our results, we present snapshots of a simulation at several points, as well as a demonstration graph that tracks statistics from several simulation runs.

3.1 Simulation Example

Each figure below shows a sample population. Viral load corresponds to node color, with lighter colors corresponding to a higher viral load. By the conclusion of the simulation, the only nodes remaining either have a very low viral load and are on ART, or were never infected.

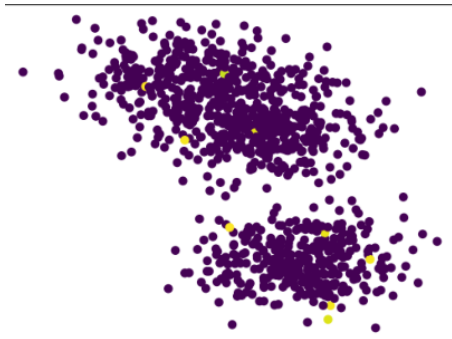


Figure 1: Week 0: Initial

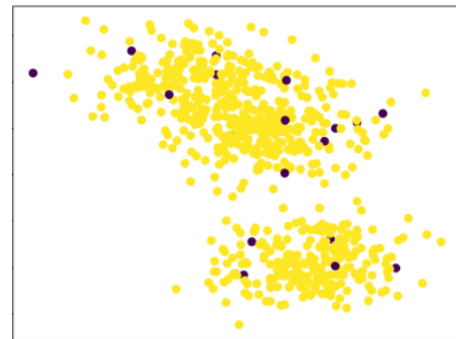


Figure 4: Week 10

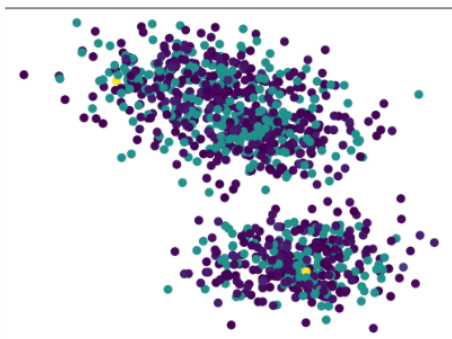


Figure 2: Week 1

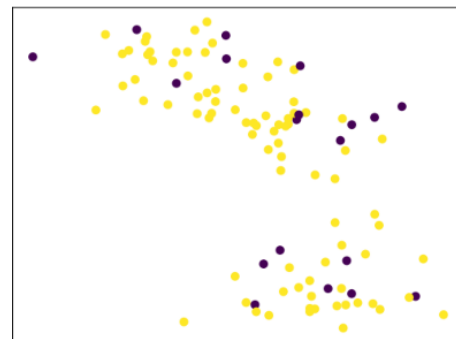


Figure 5: Week 50

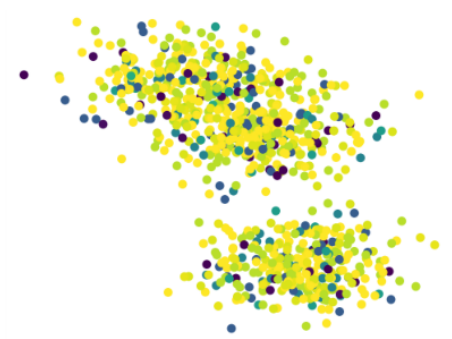


Figure 3: Week 5

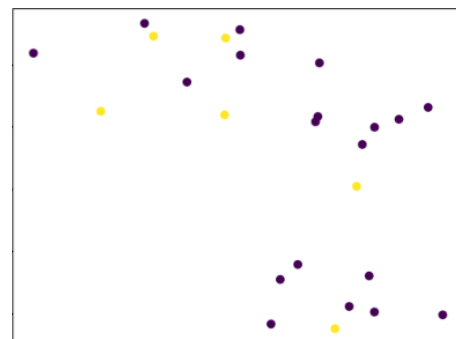


Figure 6: Week 100

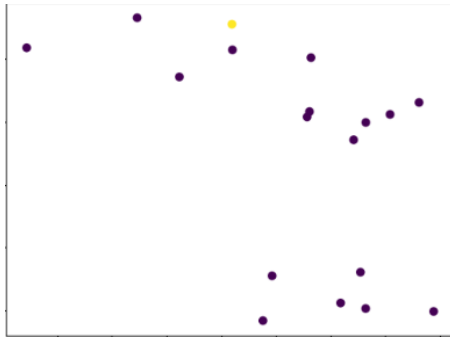
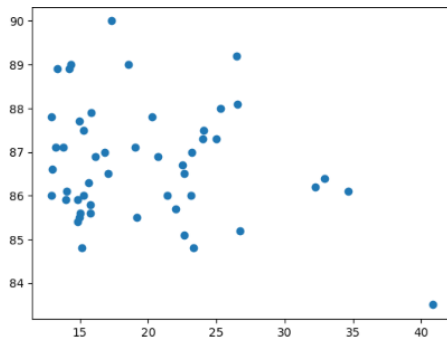


Figure 7: Week 500

3.2 Clinic Distance v Mortality

Below we show a graph showing the (extremely weak) relationship between mean clinic distance and mortality, taken over several simulations, each running for several weeks. Clinic effectiveness was set quite low for these simulations (an effective radius of 1 on a grid with a total width of 100 units, with 3 clinics total), which could explain the low correlation. However, the metrics for running such simulations and producing similar graphs can be very easily modularly defined, with a user being able to define a calculation to plot on the x and y axes of a graph in order to examine relationships easily.



4 Future Work

Throughout the project, we maintained a goal of making everything as flexible and expandable as possible, so there is considerable room for future improvement to the project. The logic for deciding the placement of clinics is a changeable function that could be swapped from its current setting of mean distance minimization or random selection to a more intelligent or perhaps learning-based strategy. There is also substantial room for improved complexity of the simulation: data from our sources tracks distinctions in transmission rates and mortality rates based on demographic characteristics such as sex, class, and age that could be incorporated into the functions that decide tick behavior for each node in the graph.

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